

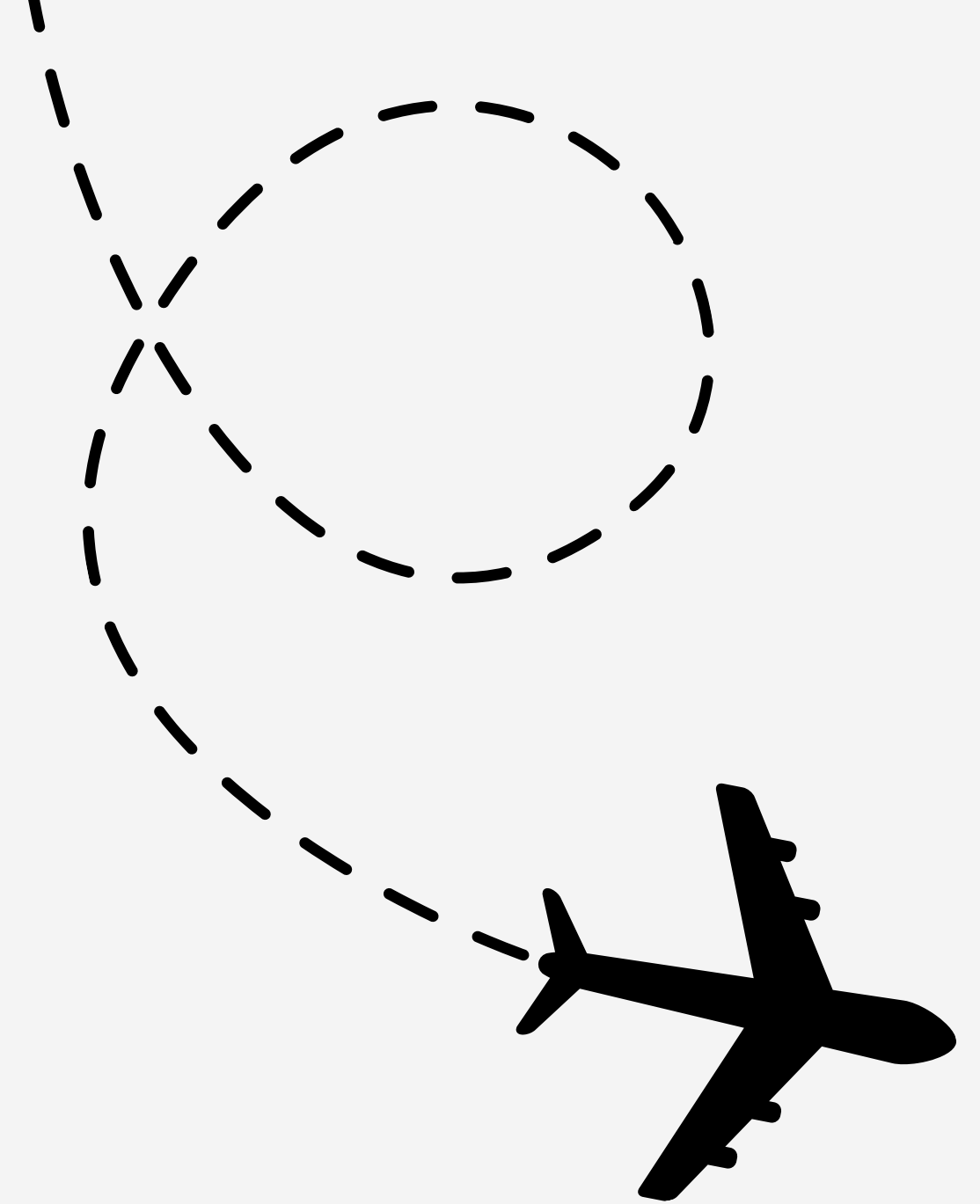
Forecasting Flight Departure Delays

Course: Forecasting and Predictive Analytics

2024-2025

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Executive Summary



Content

- ☐ Introduction
- ☐ **Exploratory Data Analysis**
- ☐ Methodology
- ☐ Results
- ☐ Discussion

Introduction

Mission

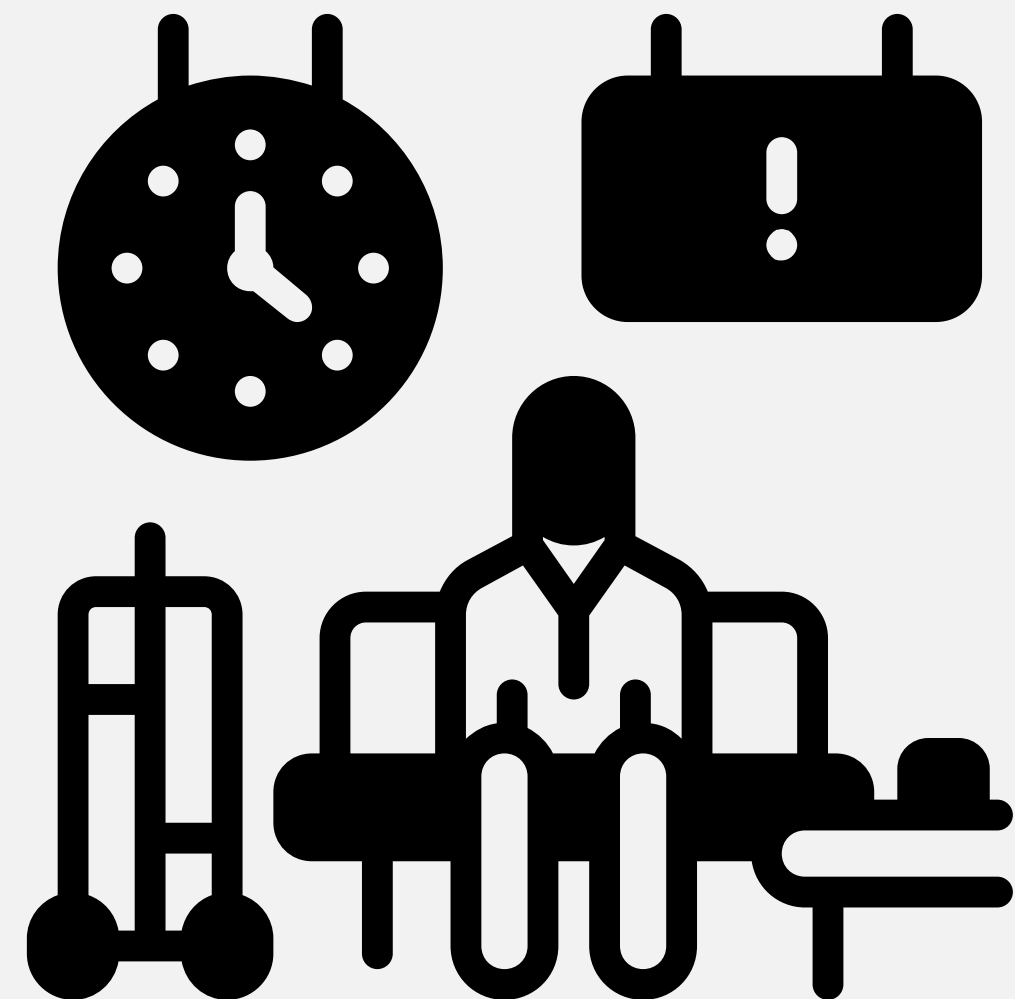
- Develop an hourly forecasting model for next-day departure delays
- Help airports and airlines anticipate operational disruptions
- Enable proactive resource planning to improve operations and passenger experience

Data Source

- **Source:** Kaggle (2019 flight data)
- **Period:** Jan 1 – Jun 30, 2019
- **Size:** 484,551 commercial flights
- **Key variables:** Date, DepTime, ArrTime, DepDelay, ArrDelay, CarrierDelay
- **Target variable:** Mean hourly DepDelay

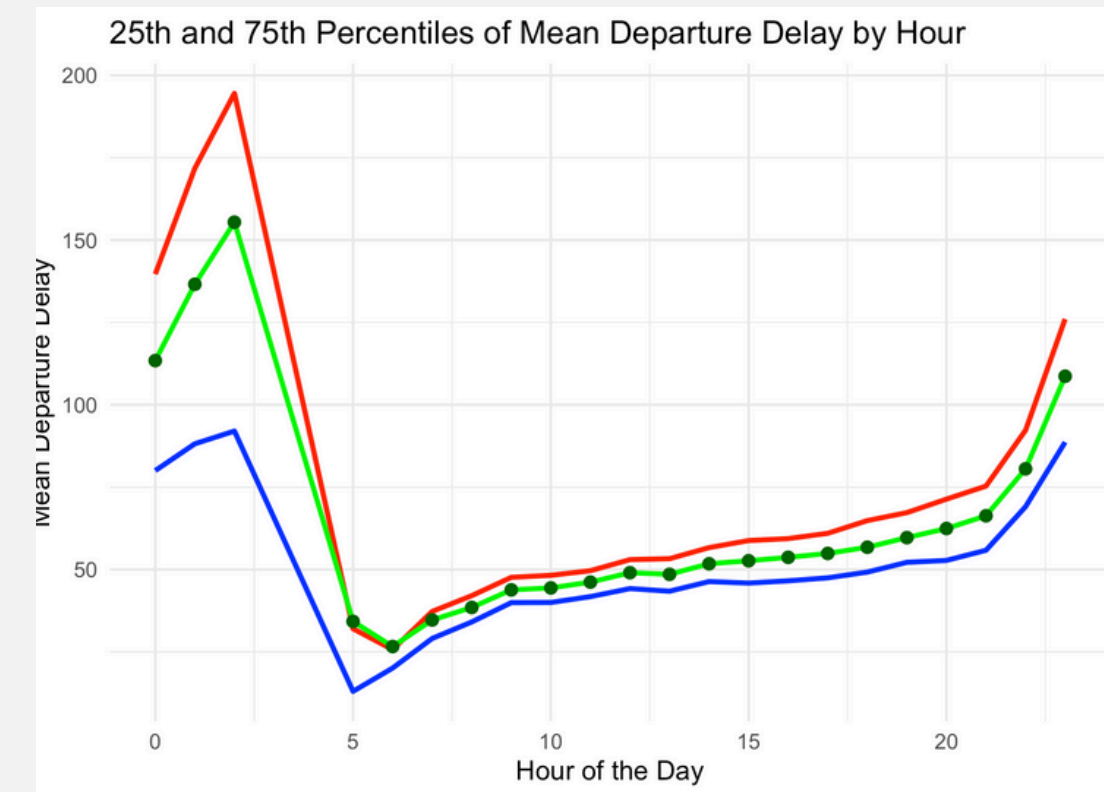
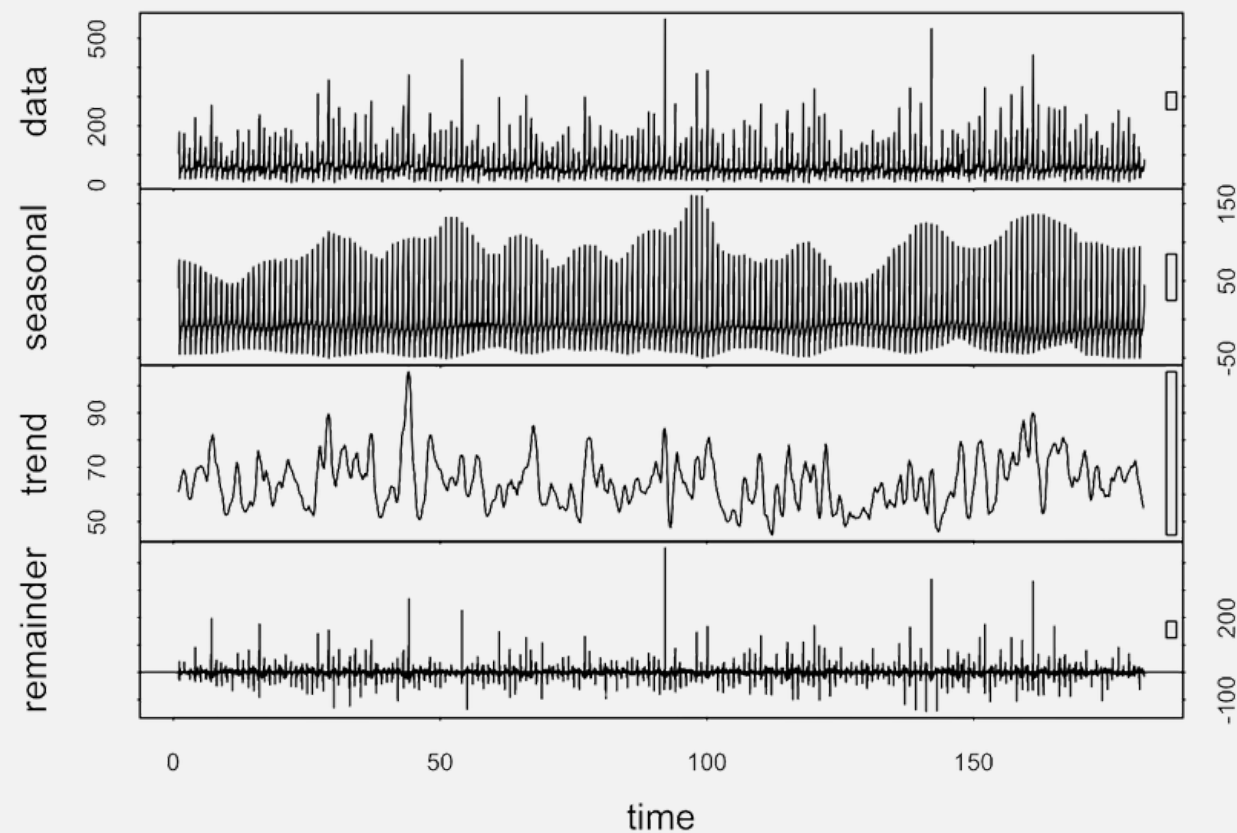
Problem Statement

Flight delays in the U.S. are a recurring challenge that disrupt airport operations and airline services. Due to the complexity and variability of air travel, there is a strong need to anticipate daily departure delays.

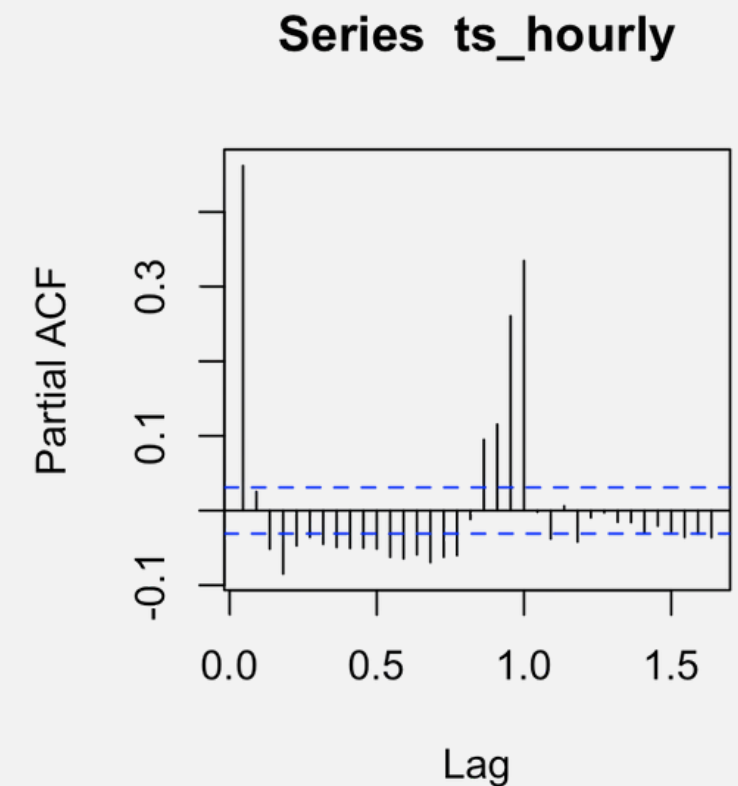
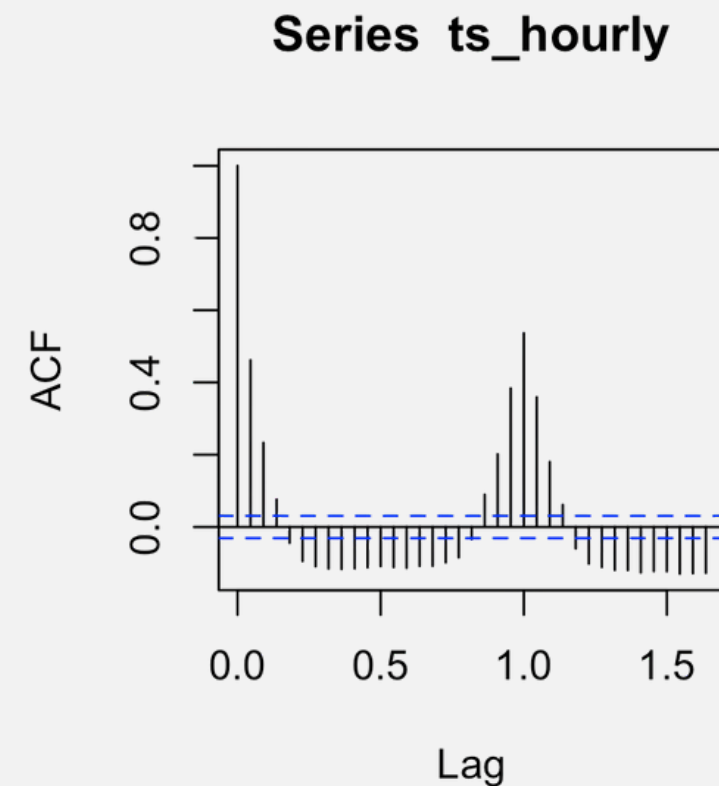


Exploratory Data Analysis

- **Data Cleaning**
 - Aggregate Flight Delays by hr
 - Handle Missing Values
- **Test for Non-Stationarities**
 - ADF Test
 - Canova Hansen Test



- **Visualize Daily and Weekly Patterns**
 - Decompose Time Series
 - Examine ACF and PACF



Methodology



Models

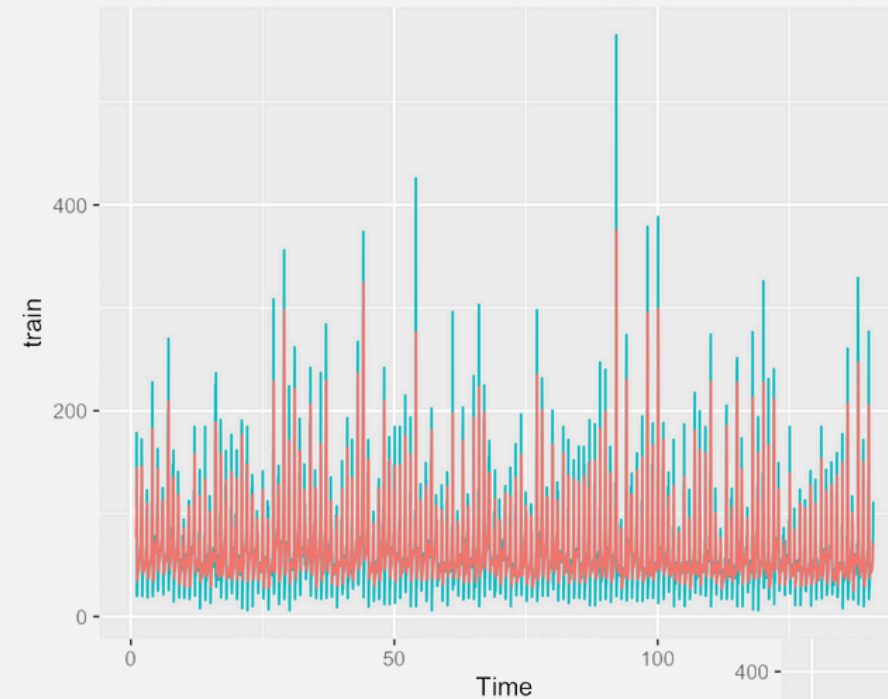
- ☐ Exponential Smoothing
- ☐ Harmonic Linear Regression + SARMA on Residuals
- ☐ GARCH Applied to SARMAX Residuals
- ☐ SARIMA
- ☐ Linear Regression Using a Covariate

Exponential Smoothing

- **Simple Exponential Smoothing (SES)**

$$\hat{y}_{t+1|t} = l_t \quad (\text{Forecast equation}) \longrightarrow$$

$$l_t = \alpha y_t + (1 - \alpha)l_{t-1} \quad (\text{Level equation})$$



series
— SES
— Train
SES (ANN)
AIC=47773

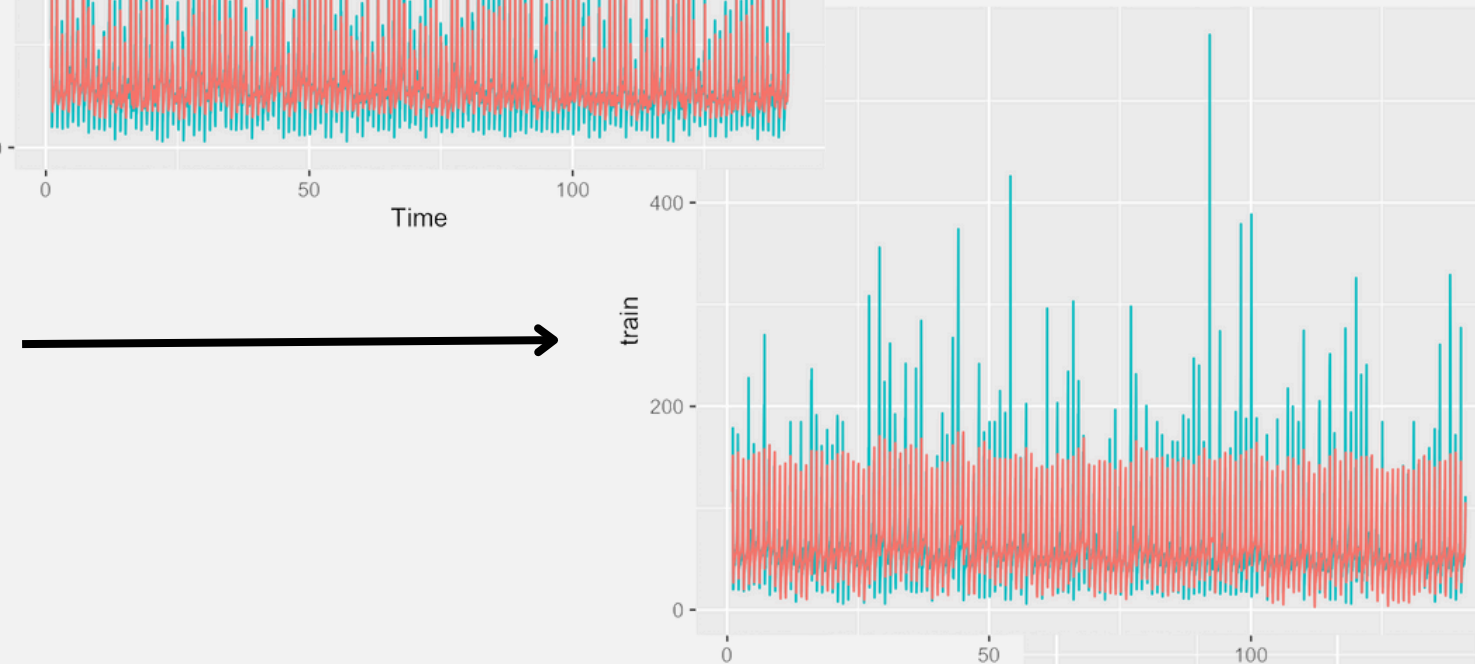
- **Holt-Winter Additive (HW)**

$$l_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1}) \quad (\text{Level equation})$$

$$b_t = \beta^*(l_t - l_{t-1}) + (1 - \beta^*)b_{t-1} \quad (\text{Trend equation})$$

$$s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m} \quad (\text{Seasonal equation})$$

$$\hat{y}_{t+h|t} = l_t + hb_t + s_{t+h-m(k+1)} \quad (\text{Forecast equation})$$



series
— HW
— Train
HW (ANA)
AIC=45451

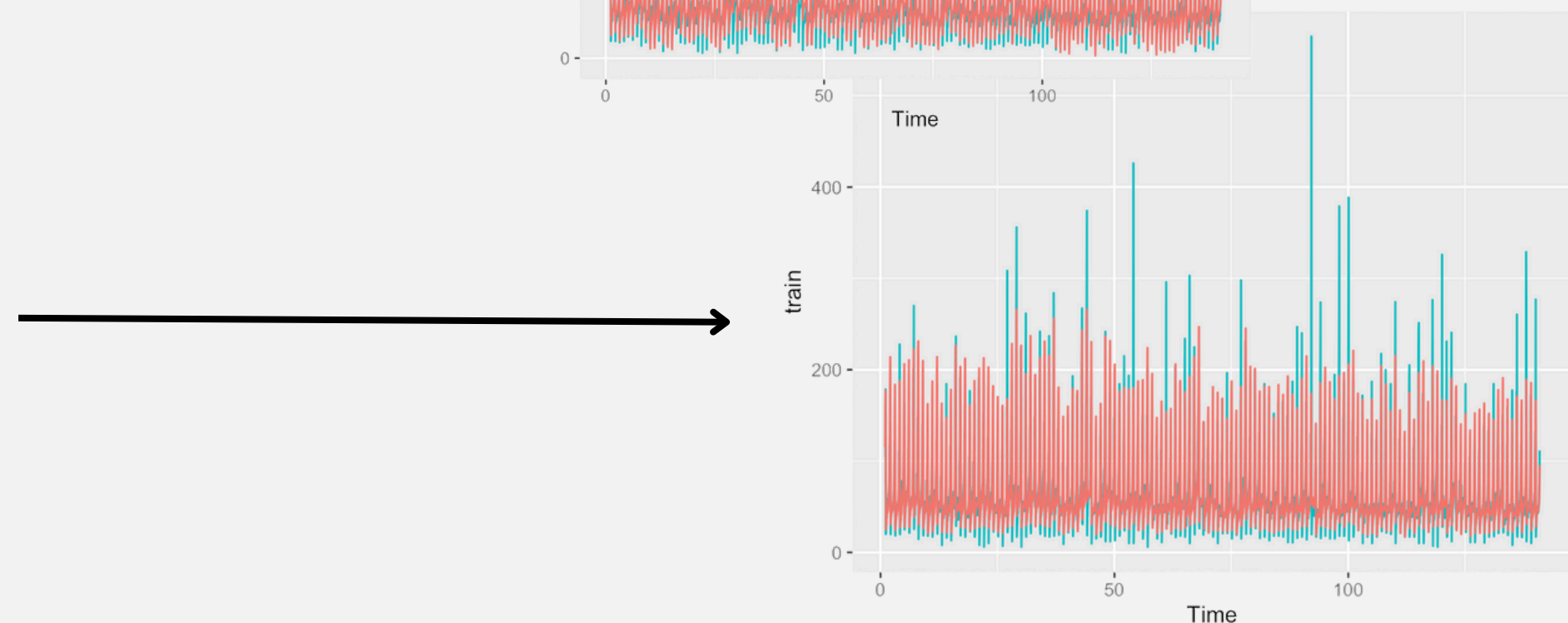
- **Holt-Winter Additive (HW)**

$$l_t = \alpha \left(\frac{y_t}{s_{t-m}} \right) + (1 - \alpha)(l_{t-1} + b_{t-1}) \quad (\text{Level equation})$$

$$b_t = \beta^*(l_t - l_{t-1}) + (1 - \beta^*)b_{t-1} \quad (\text{Trend equation})$$

$$s_t = \gamma \left(\frac{y_t}{l_{t-1} + b_{t-1}} \right) + (1 - \gamma)s_{t-m} \quad (\text{Seasonal equation})$$

$$\hat{y}_{t+h|t} = (l_t + hb_t) \cdot s_{t+h-m(k)} \quad (\text{Forecast equation})$$



series
— ETS
— Train
ETS(MNM)
AIC=42579

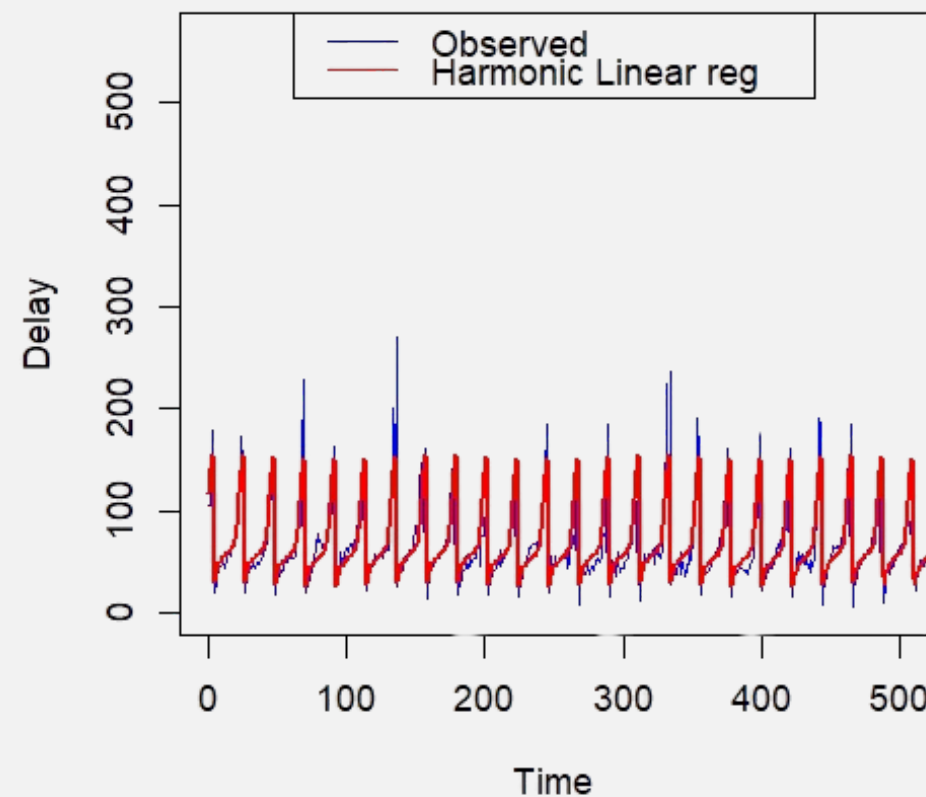
Harmonic Linear Regression + SARMA on Residuals

- Applied harmonic regression to model daily and weekly seasonality, selecting 11 and 1 harmonics via AIC.
- Model showed good fit but residuals had significant autocorrelation

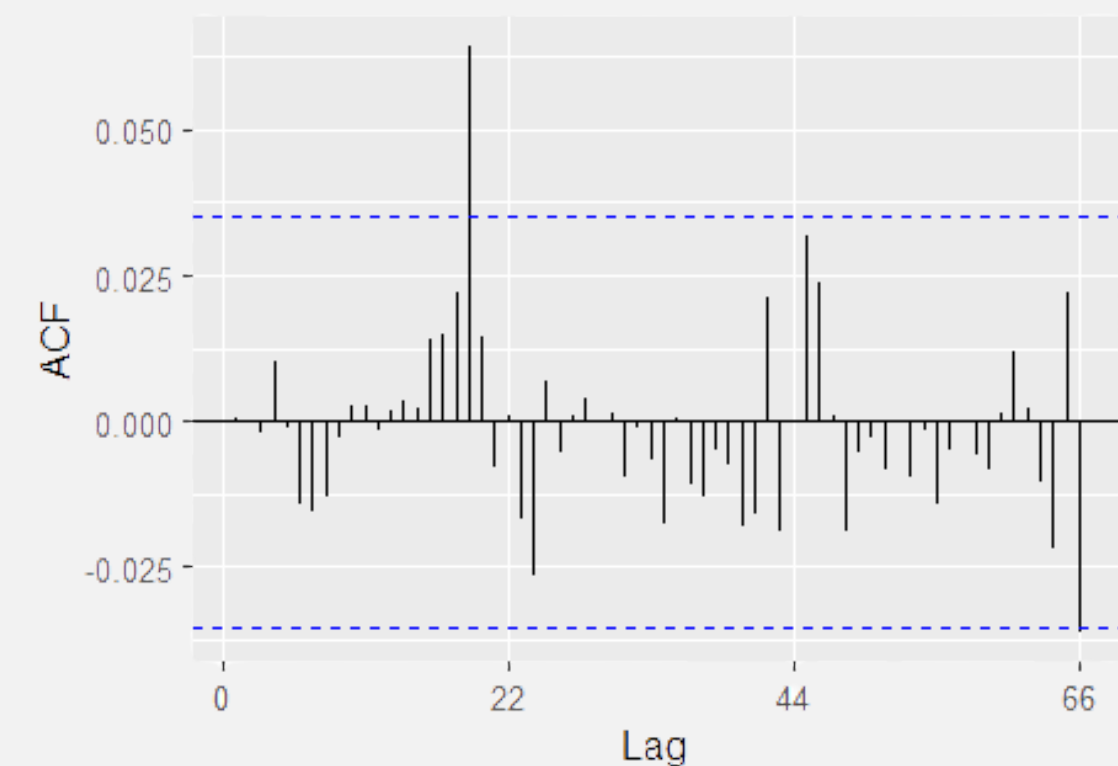
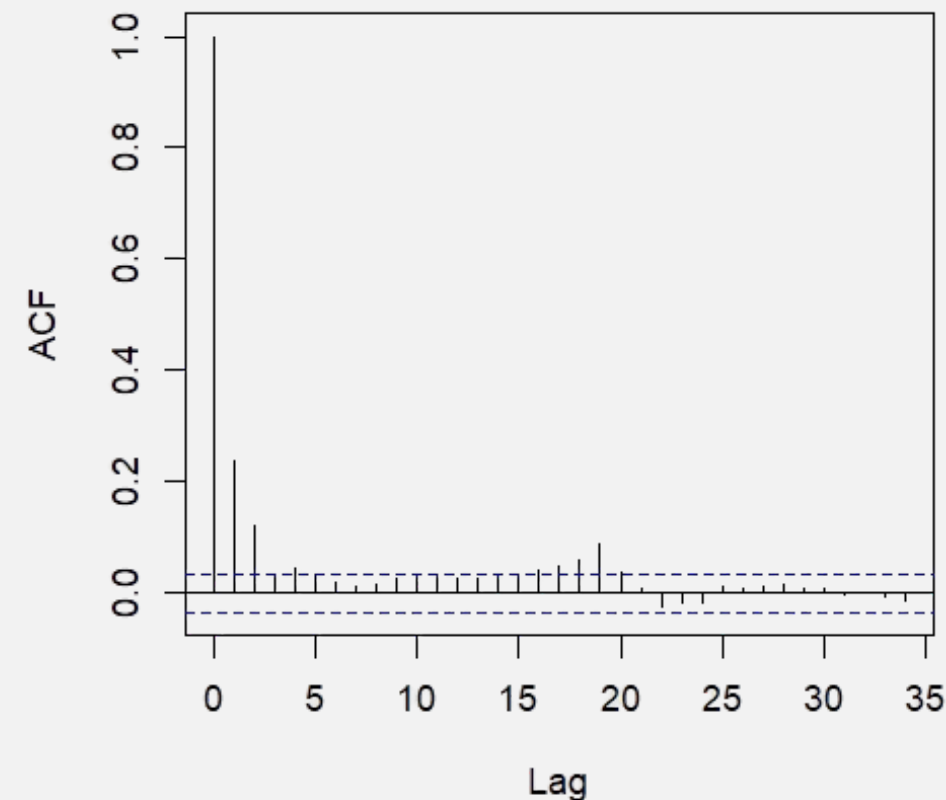
$$\longrightarrow MeandepDelay_t = \beta_0 + \beta_1 t + \sum_{k=1}^{11} \left[\alpha_k \sin\left(\frac{2\pi kt}{22}\right) + \gamma_k \cos\left(\frac{2\pi kt}{22}\right) \right] + \delta_1 \sin\left(\frac{2\pi kt}{154}\right) + \delta_2 \cos\left(\frac{2\pi kt}{154}\right) + \varepsilon_t$$

- Modeled residuals with SARMA(4,0,1)(2,0,0)[22] to capture stochastic seasonality and enhanced overall model performance.

Observed vs Harmonic Linear regression

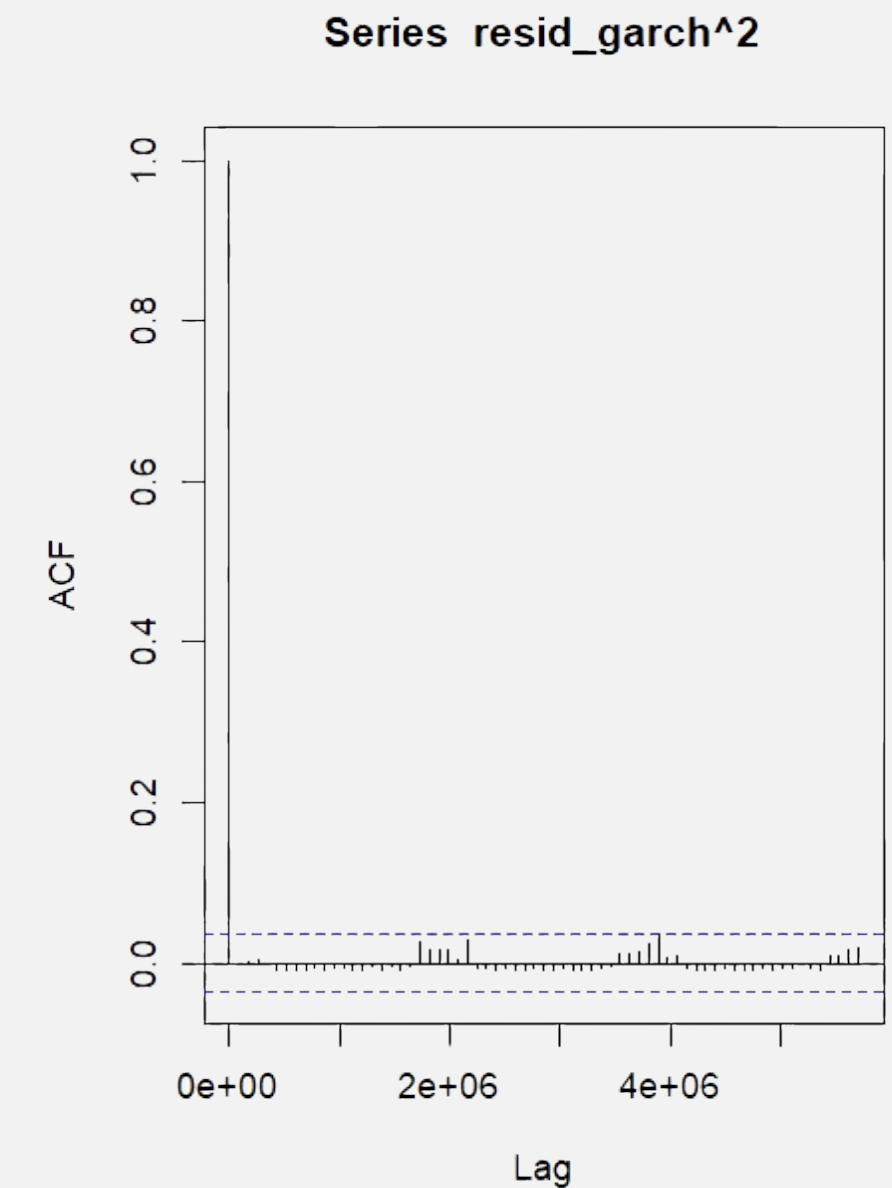
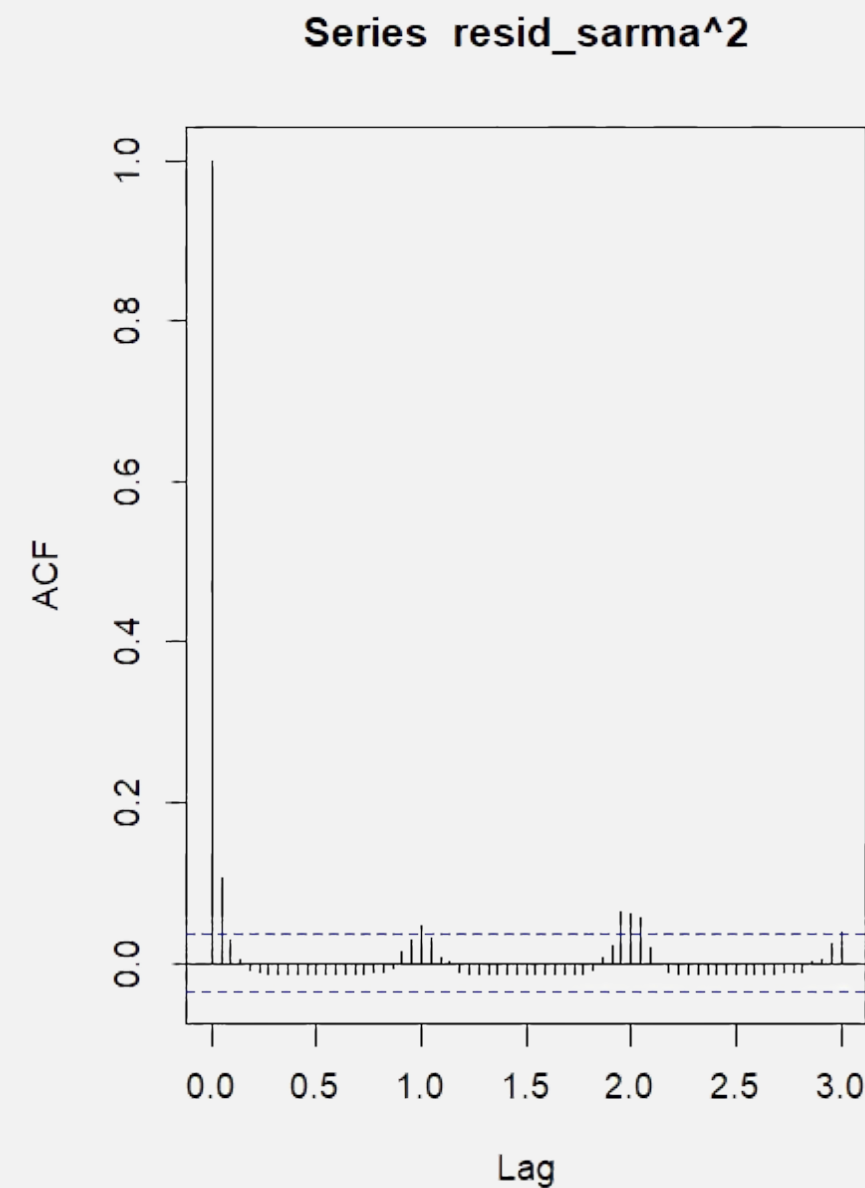


Series reg_residuals



GARCH Applied to SARMAX Residuals

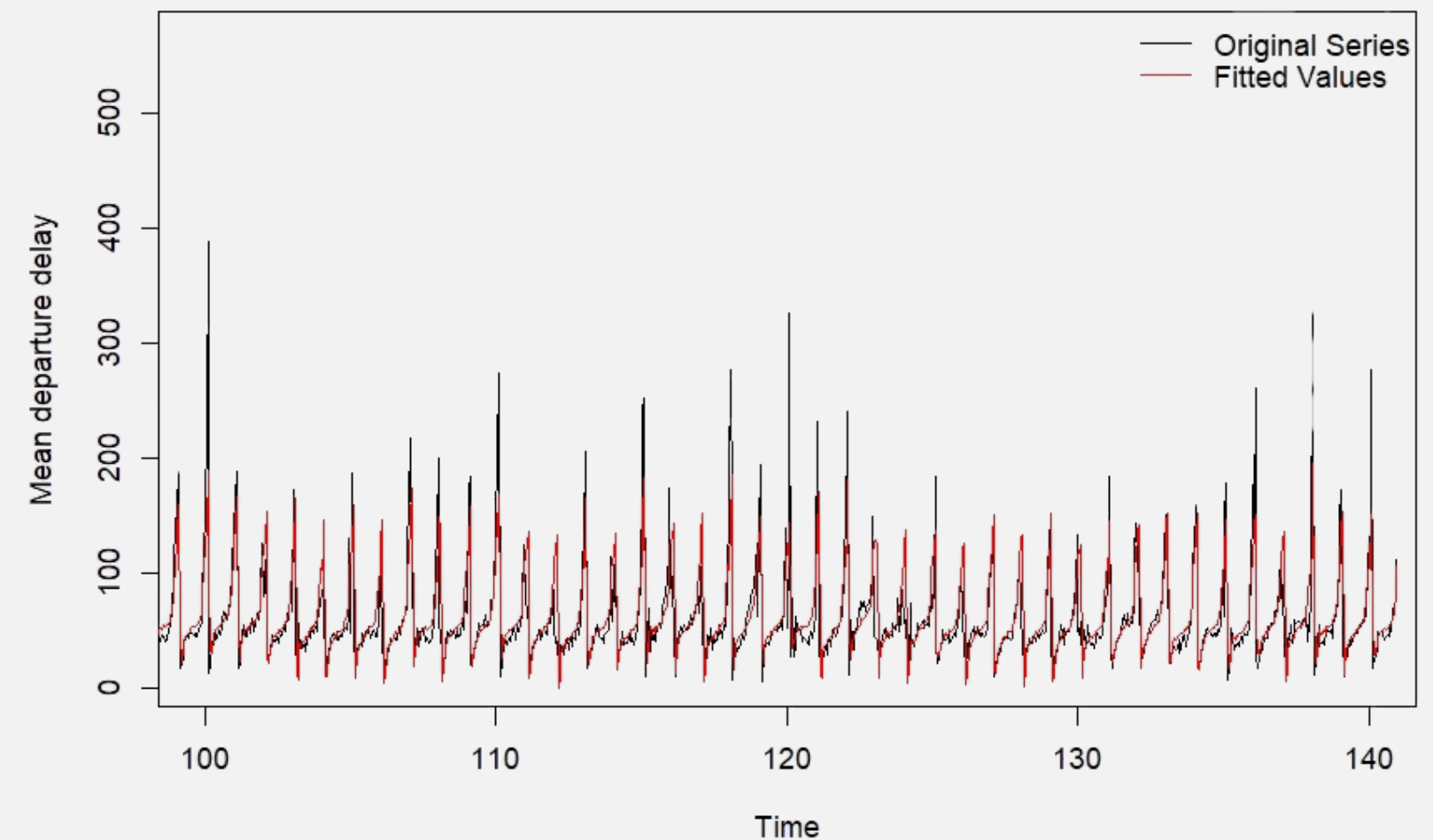
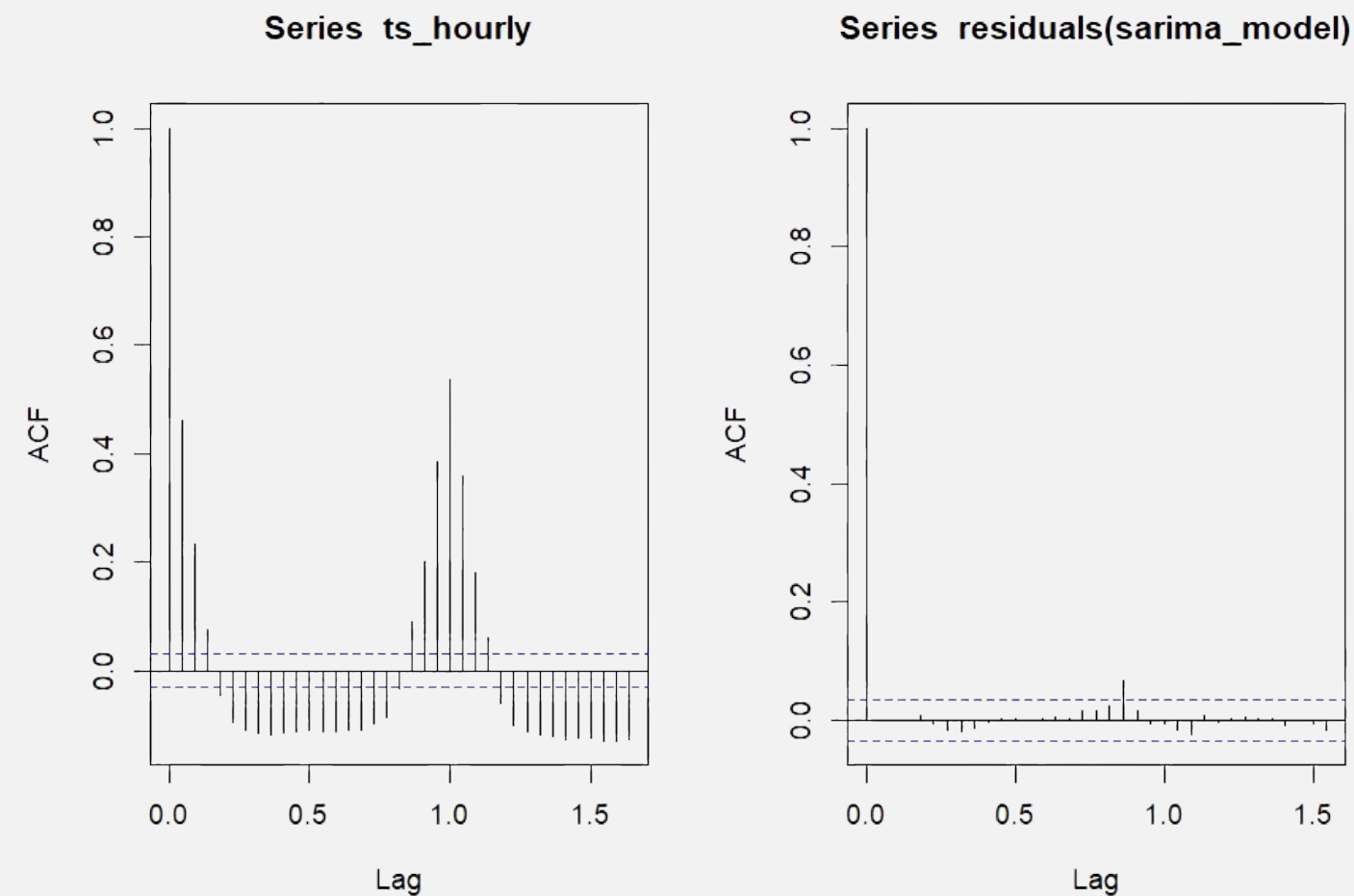
- Residuals from the LM + SARMAX model exhibited non-normality and significant autocorrelation in their squared values, indicating conditional heteroskedasticity.
- A GARCH(1,1) was used to model time-varying volatility and account for residual asymmetry.
- Following the application of the GARCH(1,1) model, the squared residuals showed no autocorrelation, indicating a satisfactory fit.



SARIMA

- Tested SARIMA as an alternative to harmonic regression + SARMA, setting $d = 0$ and $D = 1$ based on prior diagnostics.
- Selected SARIMA(4,0,1)(2,1,1)[22] using AIC as the selection criterion.

- Model effectively captured both short-term dynamics and seasonal patterns.
- Following the application of the GARCH(1,1) model, the squared residuals showed no autocorrelation, indicating a satisfactory fit.



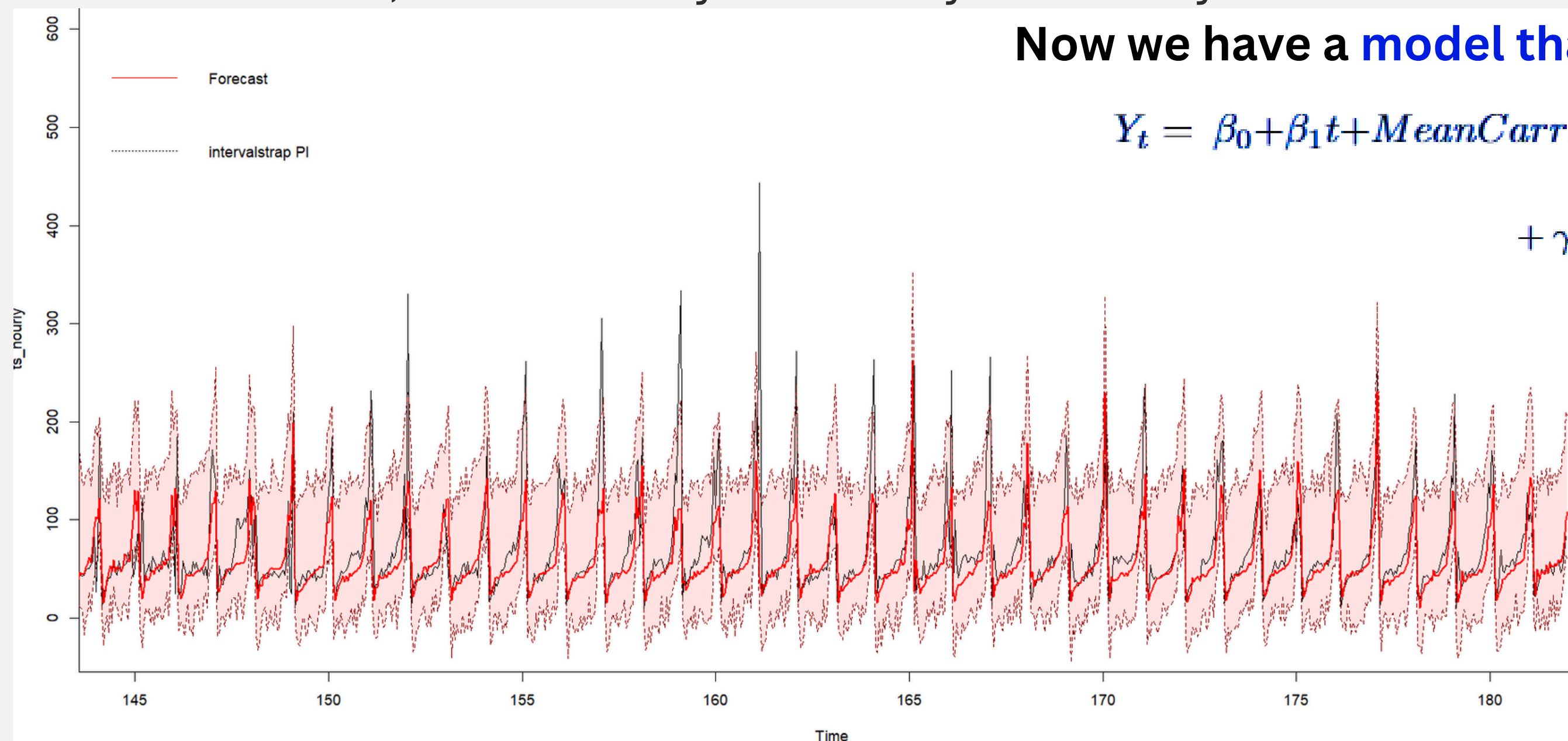
Linear Regression using a Covariate

We take into account the possibility to add a **covariate** to the model and in order to do it, we follow these steps:

- Find a covariate that has a theoretical meaning as a departure delay predictor:
we chose the (mean) Carrer Delay.
- Put it in a linear model framework: we chose a **linear model with harmonics**, like before, to handle daily and weekly seasonality.

Now we have a **model that appears like this:**

$$Y_t = \beta_0 + \beta_1 t + MeanCarrDelay_t + \sum_{k=1}^{11} \left[\alpha_k \sin\left(\frac{2\pi kt}{22}\right) + \gamma_k \cos\left(\frac{2\pi kt}{22}\right) \right] + \delta_1 \sin\left(\frac{2\pi t}{154}\right) + \delta_2 \cos\left(\frac{2\pi t}{154}\right) + \varepsilon_t$$

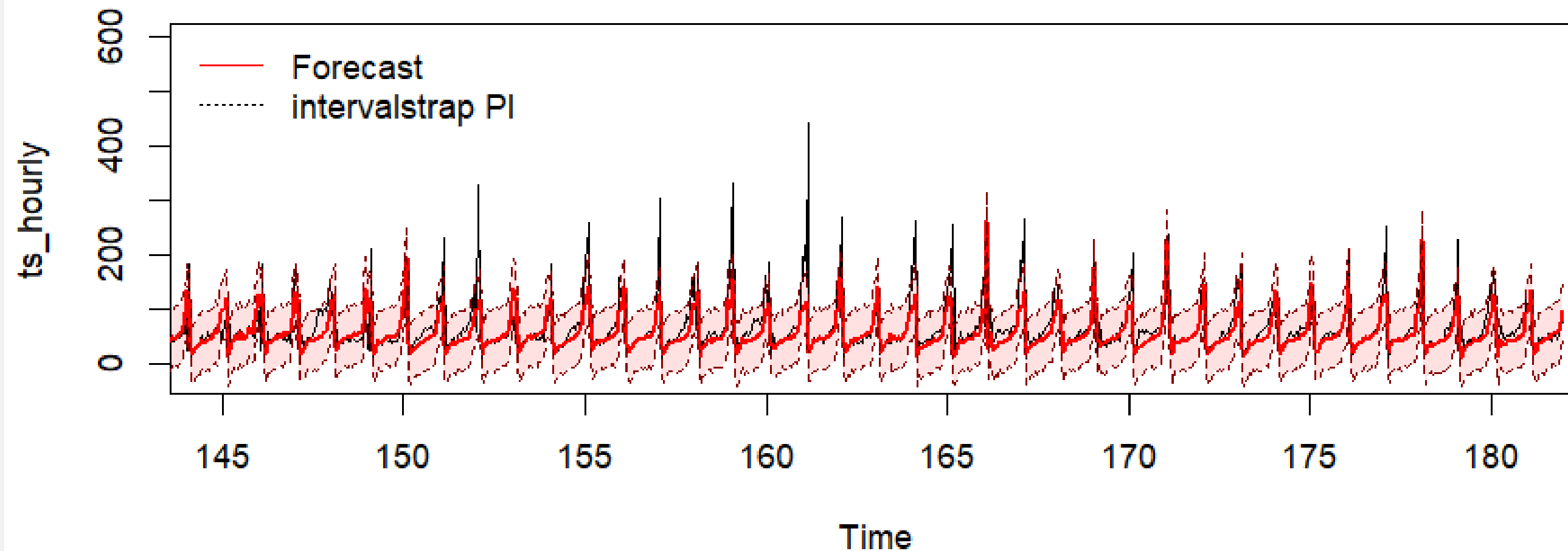


- We see that we can still **improve the fit** by modeling the residuals, so we go for a **SARMA(2,0,0)(0,0,2)[22]** on the residuals. So, now the complete model is:

$$(1 - \phi_1 L - \phi_2 L^2)y_t = \beta^\top \mathbf{X}_t + (1 + \Theta_1 L^{22} + \Theta_2 L^{44})\epsilon_t$$

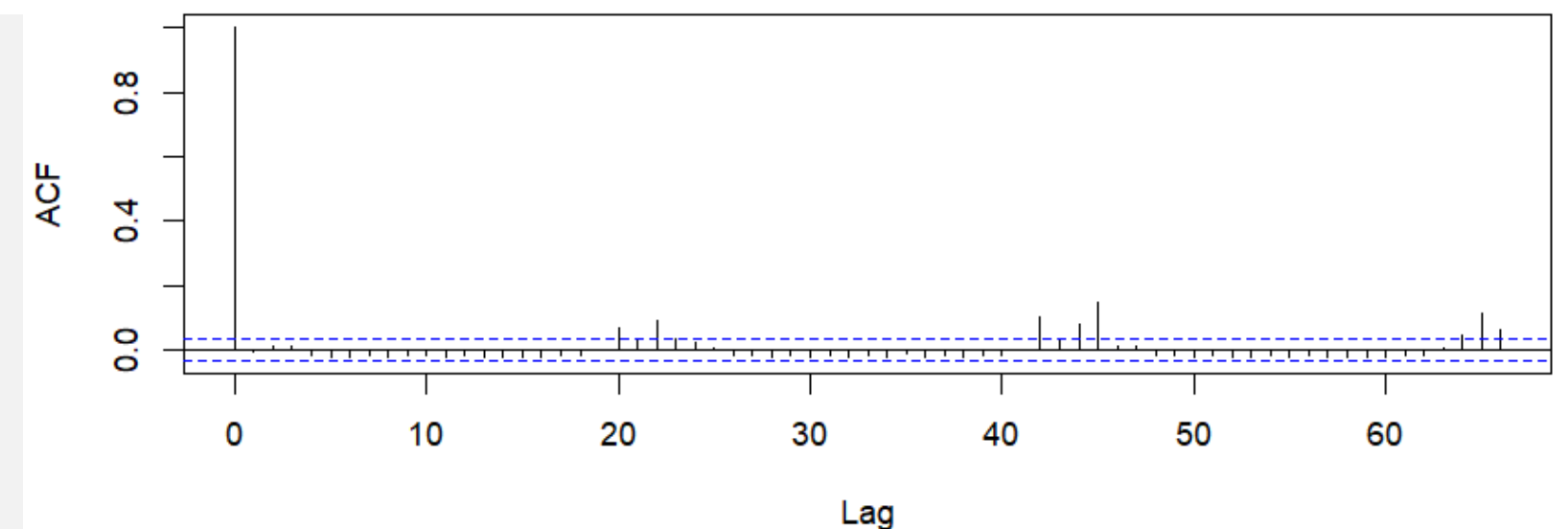
SARMAX (2,0,0)(0,0,2)[22]

with $\beta^\top \mathbf{X}_t = \beta_0 + \beta_1 t + \sum_{k=1}^{11} \left[\alpha_k \sin\left(\frac{2\pi kt}{22}\right) + \gamma_k \cos\left(\frac{2\pi kt}{22}\right) \right] +$
 $+ \delta_1 \sin\left(\frac{2\pi t}{154}\right) + \delta_2 \cos\left(\frac{2\pi t}{154}\right)$



Series resid_cov.garch^2

And we finish off by applying a **Garch(1,1) model** to manage the changing volatility of the residuals in time and the final result on the ACF of the residuals is —————>



Results

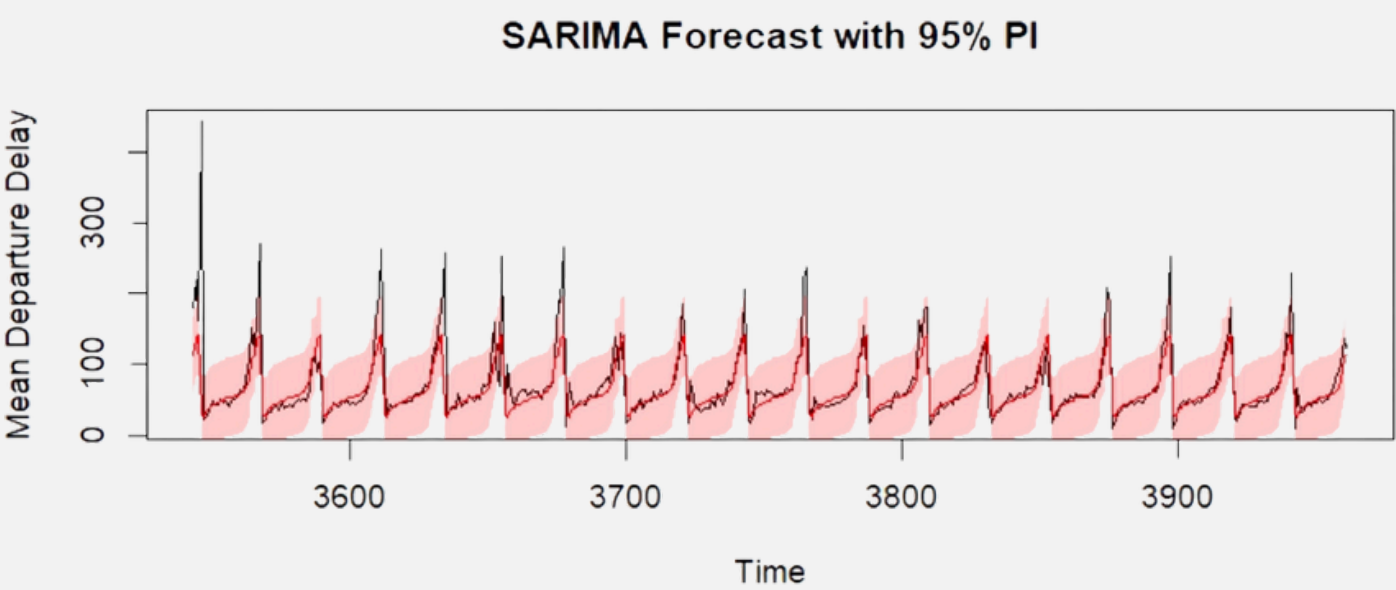
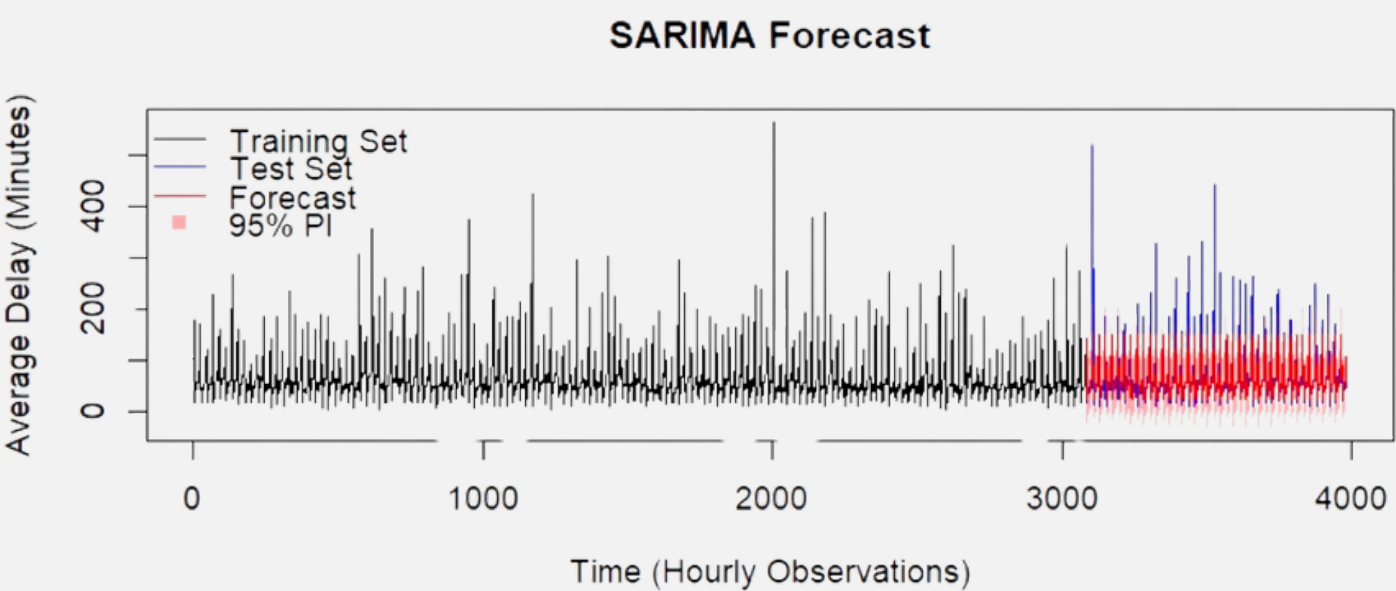
Model Performances/Evaluation

- Used a rolling forecast approach with repeated train-test splits to compare LM + SARMA + GARCH, ETS, and SARIMA.
- SARIMA achieved the best performance overall:
 - Highest unconditional coverage
 - Lowest MSE and Winkler Score

Model	Uncond. Coverage	MSE	Winkler Score
LM+SARIMA+GARCH	0.8615	3602.621	400.84
ETS	0.8994	5244.581	700.69
SARIMA	0.9567	981.526	207.99

Model	DM Test
LM+SARIMA+GARCH	LM+SARIMA+GARCH vs ETS: $t_n = -3.240$
ETS	ETS vs SARIMA: $t_n = 8.994$
SARIMA	LM+SARIMA+GARCH vs SARIMA: $t_n = 10.358$

- Diebold-Mariano test confirmed that SARIMA produced significantly more accurate forecasts than both ETS and LM + SARMA + GARCH.



Discussion



Limits and Future Improvements

Our department has been able to create a decent forecast for departure delays, but there are limits to it and much room for improvement...

- ☐ A full year's worth of more current data would be helpful to improve the value of our forecasts
- ☐ “Personalized” forecasts for each major airport could be more useful given geographical differences in US
- ☐ More experimentation on the addition of covariates is needed, which could greatly improve our forecasts

Thank you.

2024-2025

